

文字解釋題

Comparison of DC and AC Power Systems

(Difference Between Active and Reactive Power)

1. DC (Direct Current) Power System:

- **Power Type:**
 - DC systems only have **active power (real power)**.
 - All the power supplied is used for performing actual work (like running electronics, charging batteries, etc.).
 - There is **no reactive power** in pure DC systems.
 - **Voltage and Current:**
 - Both are constant over time.
 - Power is calculated simply as $P = V \times I$.
 - **Efficiency:**
 - No reactive components means **simpler power management and fewer losses** in short-distance systems.
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2. AC (Alternating Current) Power System:

- **Power Types:**
 - AC systems involve both:
 - **Active Power (P):** Actually does useful work (e.g., lighting, heating, mechanical work).
 - **Reactive Power (Q):** Oscillates between the source and reactive components (capacitors and inductors), **does no net work** but is necessary for voltage control and system stability.
 - **Apparent Power (S):** The total power flow, a combination of P and Q:
$$S^2 = P^2 + Q^2$$

- **Phase Difference:**

- Because voltage and current can be out of phase in AC, reactive power becomes significant.

- **Efficiency:**

- AC allows easy voltage transformation (via transformers) and long-distance transmission, but **reactive power can reduce efficiency** and complicate system control.

Key Differences:

Feature	DC Power System	AC Power System
Power Types	Only Active Power	Active + Reactive Power
Reactive Power	Not Present	Present (due to inductance/capacitance)
Energy Efficiency	High in short distances	Efficient over long distances (with losses due to Q)
Power Factor Concern	Not applicable	Important (affects energy losses and sizing)
System Complexity	Simpler	More complex due to Q and phase management

Summary:

- DC systems are simpler with only real power involved.
- AC systems support long-distance transmission but involve both real and reactive power, requiring careful management for efficiency and stability.

Let me know if you'd like a visual or a practical application comparison too!

Comparison of Single-Phase and Three-Phase AC Voltage (RMS Value and Power Disturbances)

1. RMS Value:

- **Single-Phase Voltage:**

- Typically, the RMS voltage for single-phase AC systems is **120V or 230V**, depending on the region.

- The power delivered is **$P = V \times I \times \text{power factor}$** , and since only one phase is used, the total power is limited compared to three-phase systems.
- **Three-Phase Voltage:**
 - Each phase has the same RMS voltage as single-phase (e.g., 230V), but the **line-to-line RMS voltage** is $\sqrt{3}$ times higher ($\approx 400\text{V}$ for 230V phase voltage).
 - The total power delivered in a balanced system is **$P = \sqrt{3} \times V_{\text{line}} \times I_{\text{line}} \times \text{power factor}$** , making it more efficient for delivering large amounts of power.

2. Power Disturbances:

- **Single-Phase:**
 - More prone to voltage dips, surges, and flickers under load changes.
 - Power flow is not as smooth, and imbalance or disturbance in the single line affects the entire load.
- **Three-Phase:**
 - Offers **more stable and continuous power delivery** due to the overlapping phases.
 - Less susceptible to disturbances; if one phase is affected, the other two can still maintain partial operation.
 - Better suited for motors and industrial loads that require steady and high power.

Summary:

- Three-phase systems have higher RMS voltage between lines and are more efficient in power delivery.
- They experience fewer disturbances and provide more reliable and smoother power compared to single-phase systems.

Conditions Required for Paralleling a Generator

To safely connect a generator to a running power system, it must match the grid in four key ways:

1. Voltage Magnitude Match

- The generator's voltage must be **equal** or very close to the grid voltage (usually within $\pm 5\%$).
- If it's too high or too low, you get circulating currents and possible tripping.

2. Frequency Match

- The generator's frequency must match the system frequency (e.g., 60 Hz or 50 Hz).
- If the frequencies differ, the generator will **try to speed up or slow down** the grid — not good!

3. Phase Sequence Match

- The generator's phase rotation (A-B-C) must be the **same as the grid**.
- If the phase sequence is wrong, it's like trying to connect a clockwise-turning motor to a counter-clockwise system — disaster.

4. Phase Angle Alignment

- The generator voltage waveform must be **in phase** with the grid voltage at the moment of connection.
- Think of it like jumping onto a moving merry-go-round — you have to match the timing exactly or get thrown off.

 Optional but Important:

5. Synchronization Devices

To make sure these conditions are met, power systems use synchronization tools like:

- **Synchrosopes:** Show the phase difference and frequency drift between generator and grid.
- **Automatic synchronizers:** Automatically adjust generator speed and voltage, and close the breaker at the perfect time.

- **Check sync relays:** Prevent closing the breaker if synchronization isn't perfect.

💥 What Happens If You Ignore These?

- **Out-of-sync closing** can cause:
 - Huge **transients**
 - **Mechanical stress** on generator shafts
 - **Circuit breaker damage**
 - Even **blackouts** in serious cases

🔄 Quick Checklist (The "Four Sync Conditions")

Condition	Must Match
Voltage	Magnitude
Frequency	Same Hz
Phase Sequence	A-B-C
Phase Angle	Close to 0°

⚡ TL;DR

To parallel a generator with a running AC system, you must match:

- Voltage
- Frequency
- Phase sequence
- Phase angle